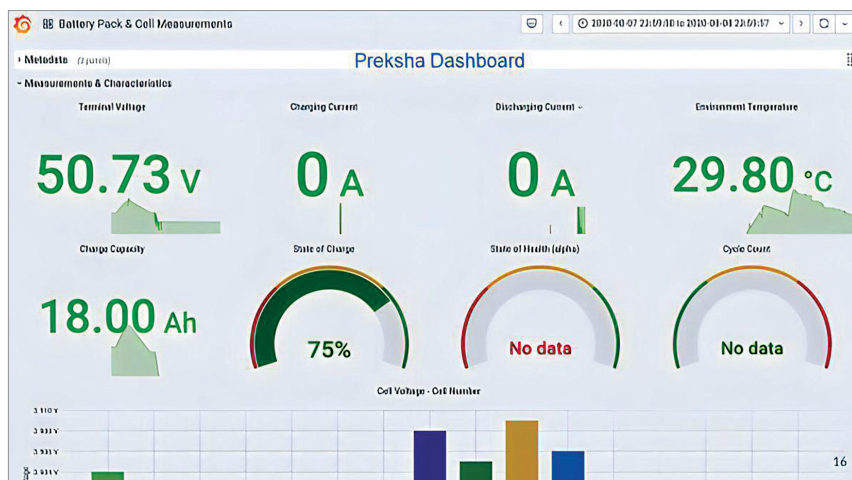


of carbon emissions, operational challenges continue to constrict its growth. In the midst of this, Gurugram based battery analytics startup iRasus Technologies is effectively wielding the power of data analytics to reduce risk, improve efficiency, and increase trust in batteries.

Founded in 2018 by Arjun Sinha Roy and Anirudh Ramesh, iRasus has designed its flagship platform named *Preksha*, which leverages the efficiency in visualisation, analysis, and predictive models of energy data in the cleantech domain for the mobility and stationary battery storage segments.

Outlining real use cases and potential risks associated with the usage of Li-ion batteries, Ramesh says that the analytics model helped identify many operational challenges. Right from warranty management to electric battery thefts and even challenges faced by battery swapping stations, iRasus has been utilising battery analytics to ensure energy storage systems are safe, reliable, and affordable. The company is currently offering its Software-as-a-Service (SaaS) to electric vehicle original equipment manufacturers, battery pack manufacturers, fleet operators and more.

The data generated from the sensors inside the battery is collected through an Internet-of-Things layer and an analytics layer and is fed back into these business processes. "The sensors present inside of the battery are either a part of the BMS or a part



iRasus Technologies' flagship product, *Preksha*

of the telematics unit, that determine the completeness and accuracy of the data. The measurements are captured from the sensors periodically and there is no large time duration when no data is reported. That is taken care of by the telematics unit," says Ramesh.

Further elaborating on the method of data processing, Ramesh explains, "The system comes with the ability to collect these measurements from the sensors and then send them over a telecom link. A 2G/3G/4G transceiver is present inside of the telematics hardware, which acts as a gateway to dispatch both electrical as well as geographical data to the backend. Since sensor data is inherently noisy, it's filtered so that it can be utilised in a format that is useful for analytics."

Explaining what differentiates them from the other battery analytics companies, Ramesh says, "There are two ways of analysing sensor data—you either look at individual cell voltages and multiply by the number of cells inside the battery pack to define the behaviour of the battery. We follow a top-down approach where we look at how the battery as a whole is behaving,

and then try to break it down into its components, reducing the error that occurs in the first method."

Roy adds that the team spent a lot of time educating customers about the value proposition offered by the company to counter challenges faced due to the lack of awareness.

With a team of 15 people comprising 10 engineers, the startup is in talks with different partners in the semiconductor and battery management system space. It started commercialising its services last year and is solely focused on the Indian market as of now. Following its angel funding round, the startup is now seeking to raise another round of seed funding.

"We're putting together a sales go-to-market strategy. We have already started selling and we intend to scale up our distribution as well. We are making some strategic partnerships in the industry," adds Roy.

The startup's future milestones involve developing a range of offerings for services other than personal mobility, such as electric buses and the renewable energy sector. "There is a product offering that we are building which functions as an energy management or a power management system on top of an ADAS system, only for the renewable energy segment," Ramesh concludes. **EFY**

—Yashasvini Razdan



iRasus team